

EasyMeals

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1. Team info

Roles

- Dean Leon - Full-stack developer
- Ayowade Owojori - Backend lead
- Dylan Peniuk - Project leader
- Noam Yaffe - Frontend lead

GitHub Repository

- <https://github.com/yaffenator/EasyMeals>

Communication Channels

- Microsoft Teams

2. Product description

Abstract

Meal prepping is hard. For any individual receiving food benefits or other financial aid, the idea of properly allocating a portion of their monthly income to groceries as effectively and efficiently as possible may seem like a daunting task. Introducing: EasyMeals – Automate your meal-prepping goals with a personalized AI nutrition expert. This website will allow you to enter the amount of money you allocate to food and groceries each month and select your physical goals (such as losing weight or building muscle) and their extent. Using this information, EasyMeals will create a month-long meal-prepping plan, showing you what groceries to buy, how much of each grocery item you will need, and some fun meals that you can prepare to satisfy your hunger.

Goal

The goal of this project is to make a website that allows people to skip the trouble of budgeting for food and the stress of finding recipes to prepare, by having it all in one place. It will allow people to enter their monthly food budget and automatically give a list of groceries to buy and recipes of what to cook week by week. It will also ask for food allergies and restrictions to let people be stress-free about what they are eating. Overall, this project would reduce the stress of wondering what you are able to afford and what you are able to make, by laying it all out month by month.

Current practice

Today, this is similarly found in meal plans like Hello Fresh, where you pay a monthly subscription, and they deliver ingredients to your house. The problem with this is that it is expensive, and it's impossible to know where that food is really coming from. Also, for lower-income families who rely on government aid programs like EBT/food stamps, they are unable to use that money for those programs.

Novelty

What makes our approach different is the introduction of an AI assistant that will webscrape products from local grocery stores to create specialized meal plans that are applicable to user requests. This approach will factor in the price of the ingredients, which is absent from existing services, and the ability to cook the meals yourself, where services like Factor_ ship the entire meal.

Effects

Everyone should care about their expenses with food. However, this is targeted more towards people with lower incomes, or multiple jobs, who worry about having enough money for food or spending the time to research what to make. This would make a difference because people would have easy access to find meals that work for their budget and schedule.

Use Cases

1. (Noam Yaffe) The user will narrow down their fitness goals through a variety of health-related inquiries.
2. (Dylan Peniuk) The user will be able to generate month-long meal plans that fit within their given budget.
3. (Dean Leon) The user will receive an itemized grocery list and curated recipes tailored specifically to their financial constraints and physical objectives.
4. (Ayowade Owojori) The user will be able to minimize their spending while achieving their health goals.

Non-functional Requirements

- Scalability: The system will be able to handle 100 meal-prep generation requests per day.
- Usability: The system should generate the grocery list and the meal plan in one minute.
- Security: All user data will be kept confidential, ensuring that each user is completely comfortable sharing information about their health habits and goals.

External Requirements (from Project Requirements Elicitation)

- The product must be robust against errors that can reasonably be expected to occur, such as invalid user input.
- The product must be installable by a user, or if the product is a web-based service, the server must have a public URL that others can use to access it. If the product is a stand-alone application, you are expected to provide a reasonable means for others to easily download, install, and run it.
- The software (all parts, including clients and servers) should be buildable from source by others. If your project is a web-based server, you will need to provide instructions for someone else to set up a new server. Your system should be well-documented to enable new developers to make enhancements.
- The scope of the project must match the resources (number of team members) assigned.

Technical approach

To build this meal-prepping website, we will use the React library combined with a Next.js framework. We chose this tech-stack as it will allow us to leverage features like automatic code splitting, server-side rendering, and more. We plan to use TypeScript (+ TailwindCSS as a replacement for HTML/CSS) for our frontend components and Python for our backend functions. We will send and receive information from our fine-tuned AI model, which will most likely be implemented via a Gemini API handler, and MongoDB (or Firebase, Supabase, etc.) as our database to store individual user information and preferences. If time permits, we also plan to implement a user authentication system using both basic email/password authentication and more advanced OAuth (such as Google and other OAuth providers).

Risks

The biggest risk of this project is making sure that we find the prices of food accurately, so we don't put people over budget. Another risk is making sure we fully exclude all items that are flagged as food restrictions as not to be liable for an allergic reaction or other response to food the user shouldn't be eating. Another risk is that our program will collect sensitive user data (weight, lifestyle, ect.), so our program should be secure enough to ensure that this data is kept private.

Additionally, add the following:

- 4+ major features you will implement.
 - A weekly grocery list that stays within the user's budget

- A full month of recipes for meals the user can cook at home
- Meals targeted towards fitness goals
- Allowing the user to add food restrictions to prevent allergic reactions
- 2+ stretch goals you hope to implement.
 - Allow the user to add family members to their meal plan (prep for a family of 4)
 - Allow a certain selection of grocery stores to purchase from nearby

The major features should constitute a minimal viable product (MVP).

Timeline

- Week 3:
 - Set up the codebase and choose tech stack (most likely React + Next.JS, TypeScript, etc.)
 - Outline UI components & different pages in the web application.
- Week 4
 - Implement UI and pages in the project
 - Make backend for user inputs for specifying meal plan requirements
- Week 5
 - Implement the Gemini API
 - Begin testing meal plan generation
- Week 6
 - Present progress and idea and midterm project presentation
 - Test grocery list and meal plan generation and fix bugs such as the generated content including items or meals the user has in their restriction settings.
- Week 7
 - Begin end-to-end (E2E) integration and ensure data flows smoothly from user input to the final displayed meal plan
- Week 8
 - Begin system testing → includes backend function tests, progress coordination with the entire team, and work combination/merging.
- Week 9
 - Finalize system testing and ensure that the base product is functional.
 - Potentially move on to implementing some of our listed stretch goals.
- Week 10
 - Finalize the project and set up the final project presentation.